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Duration	16 mins 50 secs
Grade	**15.00** out of 15.00 (**100**%)

Question **1**

Correct

Mark 1.00 out of 1.00

_____ search takes a sorted/ordered list and divides it in the middle.

a. Both (1) & (3)

b. Hash

c. Binary ✓

d. Linear

Question **2**

Correct

Mark 1.00 out of 1.00

Which of the following best describes the process of a linear search?

a. Checking each element sequentially ✓

b. Dividing the list in half repeatedly

c. Skipping every second element

d. Sorting the list before searching

Question **3**

Correct

Mark 1.00 out of 1.00

Given an array arr = {45,77,89,90,94,99,100} and key = 100; What are the mid values(corresponding array elements) generated in the first and second iterations?

a. 94 and 99

b. 90 and 100

c. 89 and 94

d. 90 and 99 ✓

Question **4**

Correct

Mark 1.00 out of 1.00

If the target element is greater than the middle element in binary search, where does the search continue?

a. At the beginning of the list
b. In the right sublist ✓
c. In the left sublist
d. In the middle of the list

Question **5**

Correct

Mark 1.00 out of 1.00

In _____ checks the elements of a list, one at a time, without skipping any element.

a. Hash search
b. Binary search
c. Both (1) & (3)
d. Linear search ✓

Question **6**

Correct

Mark 1.00 out of 1.00

In binary search, what happens if the middle element does not match the target element?

a. The search continues from the beginning
b. The search continues in the left or right sublist ✓
c. The list is sorted
d. The search stops

Question **7**

Correct

Mark 1.00 out of 1.00

If a list contains 1000 elements, how many comparisons would a binary search typically make in the worst case?

a. 500
b. 1000
c. 100
d. 10 ✓

Question **8**

Correct

Mark 1.00 out of 1.00

In which situation is linear search more efficient than binary search?

a. When the list is large and sorted
b. When the list is large and unsorted
c. When the list is small and unsorted ✓
d. When the list is small and sorted

Question **9**

Correct

Mark 1.00 out of 1.00

During a linear search, what is the maximum number of comparisons needed to find an element in a list of size n?

a. n ✓
b. n-1
c. n/2
d. log n

Question **10**

Correct

Mark 1.00 out of 1.00

Which of the following is a type of searching method?

a. Merge search
b. Bubble search
c. Linear search ✓
d. Quick search

Question **11**

Correct

Mark 1.00 out of 1.00

Which of the following is a limitation of binary search?

- a. It requires the list to be sorted ✓
- b. It can only be applied to large lists
- c. It is slower than linear search for small lists
- d. It does not work with negative numbers

Question **12**

Correct

Mark 1.00 out of 1.00

What is the first step in binary search?

- a. Compare the target element with the first element in the list
- b. Divide the list into two equal parts
- c. Sort the list
- d. Compare the target element with the middle element in the list ✓

Question **13**

Correct

Mark 1.00 out of 1.00

In linear search, if the target element is not found in the list, what is the result?

- a. An error is raised
- b. The first element is returned
- c. The last element is returned
- d. The search is considered unsuccessful ✓

Question **14**

Correct

Mark 1.00 out of 1.00

In binary search, if the target element is less than the middle element, where does the search continue?

- a. In the left sublist ✓
- b. In the right sublist
- c. In the entire list
- d. At the beginning of the list

Question **15**

Correct

Mark 1.00 out of 1.00

Which of the following scenarios is best suited for applying binary search?

a. When the list contains duplicate elements
b. When the list is unsorted
c. When the list is sorted ✓
d. When the list is very small